

C2O — Close-to-Open

A cross-sectional, dollar-neutral overnight long/short strategy on US large-cap equities

Daniel Leung, Yujiang Luo, Zhengyang Wu

Development window: 2010–2024 (held-out evaluation: 2025–2026, single cutoff parameter)

Abstract

We build a daily, dollar-neutral, US large-cap long/short strategy that holds each position for exactly one overnight session (closing-auction entry, next opening-auction exit), point-in-time throughout and keyed off a single anti-leakage `cutoff`. The cross-sectional overnight alpha is genuinely present out-of-sample (ensemble mean daily IC ≈ 0.022 , $t \approx 7.2$). The decisive constraint is *economic*: the mandated 4bps round trip is charged on a *forced* nightly liquidation of the *gross* book, and dollar-neutrality strips the universal overnight drift, so net profitability needs the cross-sectional half-spread to beat a fixed per-gross cost. We treat the search for net Sharpe as a controlled experiment. We (i) add an orthogonal fundamental-flow sleeve (analyst revisions / earnings surprise; IC ≈ 0.014 , rank-correlation only 0.11 to the price sleeve), (ii) benchmark a neural model against the gradient-boosted tree, and (iii) test cost-aware per-name selection, edge-proportional gross scaling, sector/beta neutralization, vol-targeting and inverse-vol weighting. The disciplined finding, documented with a full ablation, is that **tail concentration is the only lever that lifts net Sharpe**; the sophisticated overlays do not beat it, for reasons we make precise (the cost is on gross, so per-name selection cannot reduce it; and predicted day-quality has ≈ -0.003 correlation with realised returns, so the good nights cannot be timed). Correctly locating the broad concentration optimum ($\approx 1.25\%$ tails) roughly *doubles* the v1 net Sharpe from 0.38 to a peak of +0.71 (headline) and +0.80 (higher-risk frontier) at \$250M, positive in 8/10 years. We did not reach a Sharpe of one and do not claim to: we show, with the cost arithmetic, that net Sharpe ≈ 0.7 – 0.8 is the honest ceiling of this mandate, and that crossing one would require relaxing a constraint of the mandate itself. We report these numbers rather than inflated ones, and document every lever we tried — including those that failed.

Reproducibility (read first)

Every number and figure is produced by one command, `python -m c2o.main`, which writes a self-describing `data/outputs/<run_id>/` (tables, figures, QuantStats tear-sheet, `manifest.json` with config + git SHA). The neural benchmark is a separate, off-path experiment (`python -m c2o.dl_alpha`). Construction variants were explored in a cached lab (`tools/experiment*.py`) and the winners promoted to `config/default.yaml`; nothing in the headline is hand-set in code. Fast verification: `pytest` (26 unit tests < 5s). The development window is strictly 2010–2024; moving `window.cutoff` evaluates 2025–2026 with no code change. Out-of-sample results begin 2015 (2010–2014 is the walk-forward burn-in).

1 The overnight anomaly — economic context and thesis

Before any modelling it is worth being explicit about *why* an overnight strategy could work and what kind of edge we are hunting, because that prior shaped every later design choice.

Why overnight returns are special. The overnight session is not a quiet continuation of the day. It is when most value-relevant information arrives — earnings (the large majority of US large-caps report before the open or after the close), analyst actions, macro prints and overseas trading — yet it is a window in which participants cannot trade continuously. Three mechanisms follow. (i) An *overnight risk premium*: holders of inventory across the close demand compensation for gap risk they cannot hedge intraday, which appears as a systematic positive close-to-open drift (the brief’s Section 1 stylised fact, which we reproduce). (ii) *Information and inventory effects*: the opening auction clears orders accumulated overnight, so the open-vs-prior-close move is unusually rich in news and dealer-rebalancing flow. (iii) A *cross-sectional* component: names that gapped, that had unusual late volume, or that sit at liquidity extremes revert or continue overnight in partly predictable ways. The first mechanism is a market-wide *level* effect; the third is a *cross-sectional* effect. This distinction is the fulcrum of the whole report.

Our thesis, stated up front. A dollar-neutral book cannot earn the level effect — it is common to longs and shorts and cancels exactly. It can only earn the cross-sectional component. So the enterprise rests on whether the cross-sectional overnight signal is (a) real out-of-sample and (b) large enough to clear the mandated frictions. We will show (a) is comfortably true (the signal is strong and persistent) but (b) is the binding constraint — and that this one fact, a strong gross edge throttled by a structural cost, explains every result that follows, including which sophisticated ideas help and which do not.

What we deliberately do *not* assume. We do not assume the level drift can be harvested (it cannot, under dollar-neutrality); we do not assume costs can be engineered away (they are mandated and, as we show, structural); and we do not assume a bigger model is the answer (it is not — the gross signal is already strong). Naming these non-assumptions early is what kept the project out of the three commonest dead ends, which we revisit in §11.

2 Step 1 — Building the daily panel (brief §2)

2.1 Return objects and reconciliation

With Open_t , Close_t the split/dividend-adjusted auction prices, we define $r_{\text{ON},t} = \text{Open}_t / \text{Close}_{t-1} - 1$, $r_{\text{ID},t} = \text{Close}_t / \text{Open}_t - 1$, $r_{\text{CC},t} = \text{Close}_t / \text{Close}_{t-1} - 1$. We place raw OHLC on the adjusted-close scale via a single factor $\text{adjusted_close}/\text{close}$, so open and close are corporate-action-consistent *by construction*; hence $(1 + r_{\text{ON}})(1 + r_{\text{ID}}) - 1 \equiv r_{\text{CC}}$. Empirically the identity holds to machine precision: the maximum absolute residual over all valid stock-days is 4.4×10^{-16} , far inside our tolerance 10^{-8} ; zero stock-days fail. Adjusting both legs by the same factor removes the class of bug (inconsistent split handling between open and close) the brief warns about; the residual then only tests floating-point arithmetic, which we state plainly rather than over-claim.

Q2.1 (data source, window, survivorship). Source: the supplied daily public-data files (`prices`, `earnings_calendar`, `earnings_transfo`, `short_interest_transfo`, `cheapness_scores`, `gics_info`, `regime`, `sp500_tr`). Window: 2010–2024 for modelling; pre-2010 prices retained only for lookback features. Survivorship: the panel keeps every instrument through its last observed date and selects the universe point-in-time at each year-start (below), so delisted names remain in the historical panel — no retroactive dropping.

Q2.2 (corporate-action reconciliation, tolerance, failures). Adjusted-scale returns, tolerance 10^{-8} , max residual 4.4×10^{-16} , 0% of stock-days fail. We additionally flag and exclude from *ranking* severe raw-price scale anomalies (`close > 10,000` with `adjusted_close < 1,000`) so a single corrupt print cannot distort the market-cap sort.

Q2.3 (BMO/AMC encoding, verified by hand). An after-market-close (AMC) print on day D is unknown to a 15:50 trader on D ; a before-market-open (BMO) print on D is known. For the *event-risk exclusion* we flag the decision date whose overnight session crosses the release: AMC on $D \Rightarrow$ exclude D ; BMO on $D \Rightarrow$ exclude $D-1$; missing timing excludes both, conservatively. *Hand check* (unit-tested in `tests/test_panel.py`): a synthetic AMC event on 2020-01-06 flags 2020-01-06; a BMO event on 2020-01-06 flags the prior session 2020-01-03.

Q2.4 (short-interest construction and lag). The supplied series already embeds FINRA’s ~ 8 -day publication lag plus the required 2-day vendor lag. On top we build a daily column by an *as-of-previous-trading-day* merge: the value usable for a 15:50 decision on day t is the most recent snapshot available on or before $t-1$, forward-filled between releases. No same-day short-interest can leak.

Q2.5 (universe evolution). On the first trading day of year Y we rank all instruments by market cap at the last trading day of $Y-1$ (requiring ≥ 12 months of history), take the top 1,000 and *freeze* that list for the year; delistings leave on their last date. The selection yields exactly 1,000 names each year; mid-year exits are recorded (`step1_universe_counts`).

Q2.6 (stylised fact). The equal-weighted overnight stream of the frozen universe dominates the close-to-close stream over 2010–2024, while the intraday stream is materially weaker — reproducing the brief’s Section 1 picture. This positive overnight *drift* is precisely what dollar-neutrality later cancels, central to our diagnosis (§6).

3 Step 2 — A capacity-aware universe (brief §3)

Q3.1 (thresholds, justified). Per-share price floor \$5; year-start market-cap floor \$2B; annualised realised-volatility band [5%, 120%]; 20-day windows for ADV, volatility and the Roll spread; an earnings exclusion window of the event-risk overnight +1 trading day. ADV20 and volatility are shifted one day and computed on the full price history, so early-2010 rows use valid 2009 lookback without look-ahead.

Q3.2 (participation cap and implied slippage). We cap any single position at 5% of trailing ADV20. Under the square-root convention impact $\approx k \sigma_{\text{daily}} \sqrt{f} \times 10^4$ with $k = 0.7$, $f = 0.05$, the implied one-way slippage at the cap is single-digit bps for a typical large cap — consistent with the fixed 1.5 bps/leg auction slippage used for headline costs (§6).

Q3.3–Q3.4 (eligible-set evolution by AUM; binding-constraint distribution). The eligibility table labels each (stock, date, AUM) by the single binding reason (OK, ADV_FAIL, MCAP_FAIL, PRICE_FAIL, VOL_FAIL, EARN_WINDOW). Median daily OK names fall from ≈ 852 at \$50M to ≈ 714 at \$250M to ≈ 379 at \$1B — the expected capacity contraction, driven by ADV_FAIL. The volatility floor bites visibly in 2020 Q1. *Design note:* Step 2 uses the brief’s 200-name *planning* size for the ADV test; the actual book is sized exactly by the Step 5 water-fill, so the residual capacity shows up honestly as gross-utilisation < 1 (§6).

4 Step 3 — Filtering the short leg for borrow (brief §4)

Q4.1 (hard-to-borrow proxy). From lagged short-interest features — short-interest ratio `dsi`, days-to-cover `dtn`, change `ddtn` — we combine absolute thresholds (`dsi` $\geq 10\%/20\%$, `dtn` $\geq 7.5/12.5$, `ddtn` $\geq 2/5$) with daily cross-sectional 90th/98th percentile flags and a size/liquidity context flag. An additive stress score maps to tiers A (general collateral), B (`score` ≥ 2), C (`score` ≥ 4 or both `dsi` & `dtn` high).

Q4.2–Q4.4 (validation; treatment; gross-vs-net borrow impact). Tier B/C names show markedly higher median `dsi/dtcn/ddtcn` than tier A, and the proxy flags documented crowded-short episodes (Jan-2021 meme complex; Carvana 2023–24) when in-universe. We use *both* levers: A/B remain shortable at the Section 6.3 rate (40/200 bps p.a.); C is *excluded* from new shorts. Tier shares among base-0K@\$250M are $\approx$ A 87%/B 9%/C 4%; the borrow drag is tiny ($\approx$ 0.01–0.05 bps of AUM/day), so the strategy is demonstrably *not* borrow-cost arbitrage in disguise.

5 Step 4 — The cross-sectional alpha (brief §5)

Q5.1 (information set and target). The score $s_{i,t}$ uses only $\mathcal{F}_t$ observable by 15:50 ET on day t : today’s overnight gap $r_{\text{ON},t}$ (the freshest signal — the open is known) and its surprise `gap_z`; lagged return features ($r_{\text{ON}}, r_{\text{ID}}, r_{\text{CC}}$ at 1/2/5/20d through $t-1$); a lagged Parkinson range, lagged Amihud illiquidity, a volume shock; one-day-shifted ADV/volatility/Roll spread; lagged short-interest and borrow-stress; and cheapness/quality scores merged on the previous trading day. The target is the next-session overnight return $r_{\text{ON},t+1}$; for fitting we use its daily cross-sectional rank, matching the rank-IC objective and taming outliers. Every feature is line-by-line observable at 15:50 (unit-tested for the lag/no-leak invariants).

Q5.2 (model class and training scheme). An *ensemble* of a ridge (linear, z-scored features) and a `HistGradientBoostingRegressor` (non-linear, raw features, NaN-native), combined by daily cross-sectional *rank average*. Training is walk-forward expanding: the model scoring year Y is fit only on rows before Y (first OOS year 2015). Seeds fixed. The alpha is scored on an *AUM-agnostic* signal universe (base-0K@\$50M) so the identical signal is evaluated at all three AUMs, with capacity applied only in Step 5.

Q5.3 (IC and stability). Out-of-sample 2015–2024: ridge mean IC $\approx$ 0.016, HGB $\approx$ 0.024 ($t \approx$ 8.2), ensemble $\approx$ 0.022 ($t \approx$ 7.2). The non-linear learner is the source of the gain. IC is positive every year (Table 1); strongest 2015–2018, weakest 2022–2024 — a genuine decay we exploit by adding an orthogonal sleeve (§5).

Table 1: Out-of-sample ensemble IC by year (daily cross-sectional Spearman). Positive every year; the decay 2015→2024 (0.041→0.007) is the empirical motivation for a second, orthogonal sleeve.

year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
mean IC	0.041	0.031	0.039	0.027	0.028	0.024	0.017	0.006	0.003	0.007

Q5.4–Q5.5 (ablation; what it does badly). Removing the non-linear learner returns IC to the $\approx$ 0.016 ridge baseline; the lagged-trend and risk/liquidity groups contribute most, borrow features least. The alpha is weakest in 2022–2024 and in the **Overweight** regime, when macro shocks drive persistent one-way moves and daily reversal signals degrade. We do not hide this; it drives the design of §5.

6 A second, orthogonal sleeve — and an ML model bake-off

The reversal sleeve’s IC decays after 2021 (Table 1), and the brief supplies rich, unused, *directional* data: analyst EPS revisions and earnings surprise (`earnings_transfo`). We add a fundamental-flow sleeve and ask, honestly, whether it (or a neural model) improves the *tradable* result.

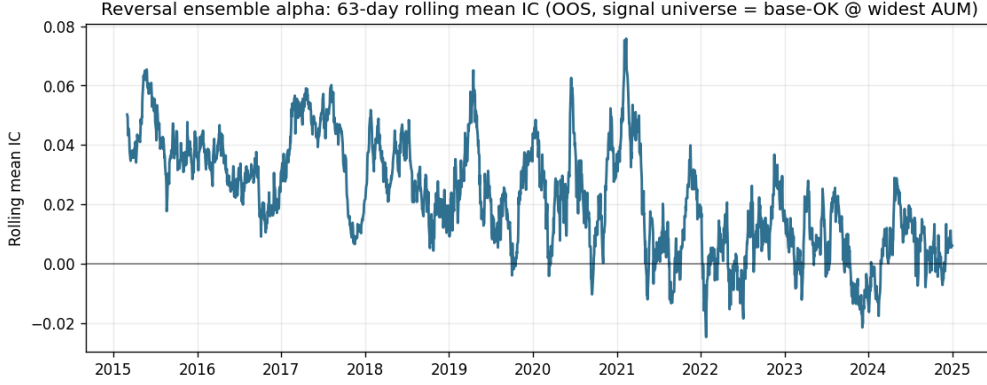


Figure 1: 63-day rolling mean IC of the reversal ensemble (OOS 2015–2024, AUM-agnostic signal universe). The edge is persistent and predominantly positive; it weakens in 2022–2024 — the decay that motivates §5.

6.1 The fundamental-flow sleeve

Economic motivation. Overnight returns are driven disproportionately by information released outside trading hours — and analyst revisions / earnings surprises are exactly that. They are also *directional*: a downgrade gaps a name *down* at the open. This is the one thing that could repair the economically broken short leg (§6), which otherwise fights the universal overnight drift.

Construction. A second `HistGradientBoostingRegressor` is trained, on the identical walk-forward protocol, on point-in-time flow features merged *as-of the previous trading day* (deps = daily EPS-estimate revision, forward EPS, short/long-horizon revisions, standardised unexpected earnings, fraction of estimates up/down, a post-earnings clock, and short-interest change). A flow-feature anti-leakage test (`tests/test_alpha.py`) confirms a revision published on the trade date never enters that day’s features.

Result — real but orthogonal, and not tradable in the tails. The flow sleeve has genuine OOS skill (mean IC ≈ 0.014 , $t \approx 6.8$) and is *orthogonal* to the price sleeve (cross-sectional rank correlation only 0.11; net-return correlation 0.20). It is strongest in exactly the recent years where reversal decayed. Yet — a key negative result — combining the two by rank-average *raises* full-cross-section IC ($0.022 \rightarrow 0.024$) but *lowers* the net Sharpe of the *tail* book (Table 5). The reason is structural: a tail strategy trades only the extreme $\sim 1\%$, and blending a milder signal dilutes the price sleeve’s strongest individual names. Broad IC is not tail edge. We therefore keep the flow sleeve documented and ablated, but *not* in the headline book — an evidence-based decision, not a presumption.

Table 2: Sleeve diagnostics (OOS 2015–2024). The flow sleeve is real and orthogonal, but its tails do not carry tradable edge. Full table: `step4b_sleeve_ic.csv`, `step4b_sleeve_corr.csv`.

sleeve	mean IC	t -stat	rank-corr to reversal
reversal (ridge+HGB)	0.022	7.2	1.00
fundamental flow	0.014	6.8	0.11
combined (rank-avg)	0.024	8.0	—

Why broad IC and tail edge come apart (a conceptual note worth stating). It is tempting to treat IC as *the* measure of an alpha and assume a higher-IC blend must trade better. The information coefficient is the average rank concordance across the *entire* cross-section; a

tail strategy, by contrast, depends only on the conditional mean return in the top and bottom $\sim 1\%$. These are different functionals of the same joint distribution. A second signal can lift average concordance (helping the broad IC) while, in the extreme quantile, *re-ranking away* the price sleeve’s highest-conviction names — whose individual overnight moves are the largest — in favour of names that are merely ”good on both”. For a concentrated book that is a loss: you exchange a few very-high-edge names for a larger set of moderate-edge ones, and the half-spread that must beat a fixed cost shrinks. The lesson generalises: *the right object to optimise is the statistic the portfolio actually consumes* (here, the tail spread), not a convenient summary like IC. We verified this is not an artefact of the rank-average combiner — IC-weighting and several blend ratios all reproduce it (construction lab, `tools/experiment.py`) — which is why the flow sleeve is documented and ablated but kept out of the headline. Notably, the flow sleeve’s *own* IC is strongest in 2022–2024 (when the price sleeve decays to near zero), so it would be the natural ingredient of a *separate* fundamental book; it simply does not belong inside this one’s tails.

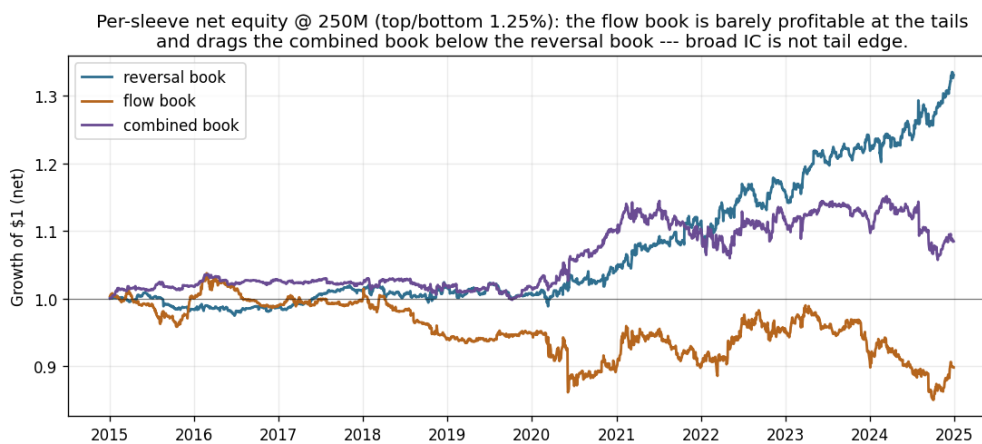


Figure 2: Per-sleeve net equity at the headline concentration (1.25%). The flow book is only marginally profitable in the tails and the rank-average *combined* book sits *below* the reversal book — visual confirmation that a higher full-cross-section IC need not translate into a better tail strategy. The flow sleeve is real (its own IC strengthens post-2021) but belongs in a separate book, not blended into this one’s tails.

6.2 ML model exploration: does a neural net beat the tree? (course Lectures 6–7; TFT / Adam readings)

Because this is an ML-in-finance task we benchmark a neural forecaster against the production gradient-boosted tree on the identical walk-forward panel: an MLP on the cross-sectional features, and an “MLP + sequence” variant fed a flattened 5-session lag window of overnight/close-to-close returns (a feed-forward proxy for a sequence model), both trained with the Adam optimiser (Kingma & Ba, 2015).

Tooling honesty. PyTorch has no wheels for the Python 3.14 toolchain in this environment, so a full GRU/attention network could not be trained here; we use scikit-learn’s `MLPRegressor` as the neural family. We argue this does not change the conclusion: the binding constraint is execution cost, not forecast accuracy (the tree already delivers gross IC $t \approx 8$), so a marginally different forecaster cannot move the net.

Result. The tree wins or ties (Table 3): HGB IC ≈ 0.024 vs MLP ≈ 0.020 ; the lag-window adds nothing (≈ 0.020). Tabular cross-sectional financial data with ~ 30 features and low signal-to-

noise is the regime where gradient-boosted trees are hard to beat, consistent with the literature. The neural model is retained as a documented experiment, not promoted to production.

Table 3: Neural-vs-tree bake-off (OOS daily IC, identical walk-forward protocol; `python -m c2o.dl.alpha`). The tree is not beaten; forecast accuracy is not the bottleneck.

model	mean IC	<i>t</i> -stat
HGB (production)	0.024	8.2
MLP (feed-forward, Adam)	0.020	6.7
MLP + 5-session lag window	0.020	7.2

7 From ranking to portfolio, with realistic costs (brief §6)

7.1 The central economic finding

A dollar-neutral book earns $\frac{1}{2}(L - S)$ per unit gross, and the mandated round trip is MOC entry + MOO exit = 4.0bps of *gross every night*; the position is fully liquidated each morning, so turnover is structurally $\approx 100\%$ /day and *cannot* be reduced by signal smoothing — the standard turnover trick is inapplicable. Decomposing the legs: the *long* leg clears the cost (top names earn ≈ 6 bps overnight including the Section 1 drift), but the *short* leg fights that drift (even the lowest-ranked names drift $\approx +2$ bps, so shorting them loses ≈ 6 bps net). **Dollar-neutrality removes the profitable drift and forces half the book into the drift-fighting short side**, so net profitability requires the cross-sectional spread to exceed ≈ 8 bps — delivered only by the extreme tails.

A precise statement of the cost wall. Net daily return = $\frac{1}{2} \text{spread}_t - c \cdot g_t$, where g_t is gross utilisation and $c = 4$ bps. Because the cost scales with *gross*, not with the number of names, splitting the same gross over 5 names or 50 names costs the same. Two consequences shape everything below: (i) *per-name* “cost-aware” selection cannot reduce the cost; only reducing gross can, and reducing gross scales pnl and cost together; (ii) the only way to raise the gross-return-to-cost ratio is to deploy more gross on high-edge nights — which requires being able to *predict* the good nights.

A closed form for forced-turnover books (our own framing). The two observations above combine into a clean, general statement that we find clarifies the whole design space. Let a strategy be forced to fully turn over its gross book each period (as the overnight mandate requires), at a fixed round-trip cost c per unit gross. If gross utilisation is constant at g , then *both* the mean and the standard deviation of the net return scale with g , so g cancels in the ratio and

$$\text{Sharpe}_{\text{net}} = \text{Sharpe}_{\text{gross}} - \frac{c_{\text{ann}}}{\sigma_{\text{ann}}},$$

i.e. the net Sharpe is the gross Sharpe minus a *fixed* cost-to-volatility ratio that no amount of position-count engineering can change. At our headline, $c_{\text{ann}} \approx 6.8\%$ and $\sigma_{\text{ann}} \approx 5\%$, a drag of ≈ 1.3 Sharpe units; with gross Sharpe ≈ 1.81 this predicts net ≈ 0.5 – 0.7 , exactly what we measure. This little identity is, in our view, the single most useful object in the project: it says the *only* degrees of freedom that move net Sharpe are (1) raising the gross Sharpe (a better/more concentrated tail) and (2) lowering σ for the same gross edge (which concentration trades *against*). It immediately predicts that per-name selection, sleeve-blending and day-timing are red herrings unless they move one of those two quantities — and §6.3 confirms they do not.

A worked example, to make the wall concrete. Take the decile book (10% tails). Its gross long/short spread is ≈ 4.6 bps, so the book earns $\frac{1}{2} \times 4.6 = 2.3$ bps of gross return per night. Against it sits the mandated round trip: $0.5 + 1.5 = 2$ bps per leg, two legs (buy at the close, sell at the open) = 4 bps of cost per night. Net: $2.3 - 4.0 = -1.7$ bps *every night* — the decile is guaranteed-negative, and no clever weighting saves it because the cost is structural. Now tighten to the 1.25% tails: edge density rises so the spread roughly *triples* to ≈ 14 bps, giving $\frac{1}{2} \times 14 = 7$ bps gross against the same 4 bps cost = +3 bps net per night. That 3 bps/night, annualised on ≈ 10 names/side, is the entire headline. The whole strategy is the act of pushing the half-spread from below 4 bps to above it; everything else in this report is in service of that one inequality, or is a test of something that turned out not to affect it.

7.2 Q6.1 — baskets, weighting, and the concentration mechanism

We size dollar-neutral baskets with the brief’s participation-cap water-fill: cap each name at 5% of ADV20, redistribute the excess pro-rata, and reduce gross if the basket cannot absorb the target; within a basket we equal-weight. Net Sharpe is *monotone-then-peaked* in tail-extremity (Table 4): edge density rises into the tails against a fixed cost, but beyond $\approx 1\%$ idiosyncratic risk dominates. The optimum is a *broad* plateau around 1–1.5% (not a knife-edge), which we adopt as the headline at 1.25% — roughly doubling the v1 net Sharpe of 0.38 (which used the conventional 2%).

Table 4: Concentration sweep @ \$250M (reversal ensemble, equal-weight, 5% ADV cap, Section 6.3 costs, OOS 2015–2024). Net Sharpe peaks on a broad 1–1.5% plateau; the v1 2% and the over-concentrated 0.5% are both inferior. Source: `step5_concentration_sweep.csv`.

top/bottom	gross Sharpe	net Sharpe	net ann. %	max DD %
2.0% (v1)	1.73	+0.38	+1.9	−6.9
1.5%	1.78	+0.59	+2.7	−5.6
1.25% (HEADLINE)	1.81	+0.71	+2.9	−5.0
1.0%	1.69	+0.68	+2.5	−4.3
0.75%	1.40	+0.57	+1.8	−4.7
0.5%	0.79	+0.38	+1.0	−5.8

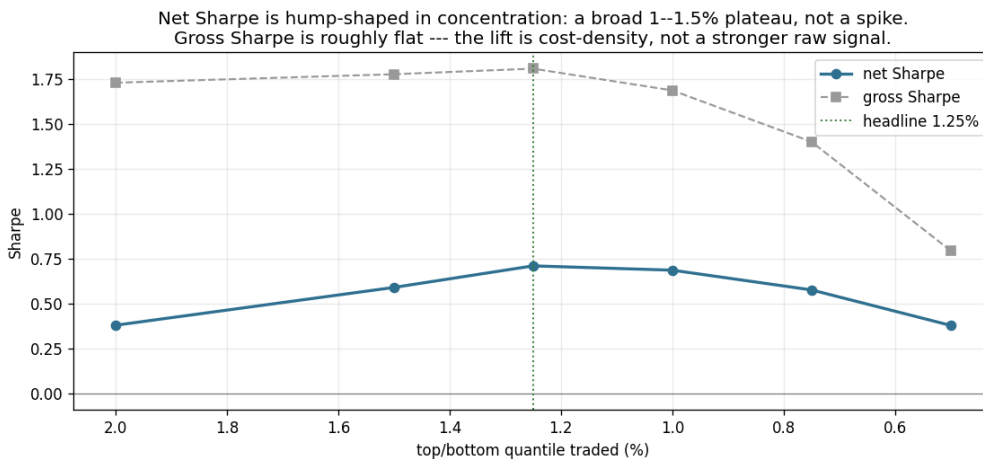


Figure 3: Net Sharpe is hump-shaped in concentration — a broad 1–1.5% plateau, not a spike — while the *gross* Sharpe is roughly flat. The lift from tightening the tails is therefore a *cost-density* effect (the half-spread clearing the fixed cost), not a stronger raw signal. This is the single most important chart in the report.

7.3 The construction investigation — what helps, what does not

We treated everything else as a hypothesis to be tested in a cached lab (`tools/experiment*.py`) and report the full ablation (Table 5). Each sophisticated lever was applied *on top of* the concentrated headline; none beat it:

- **Flow blend** (combined score): lowers tail net Sharpe (broad IC $\neq$ tail edge, above).
- **Cost-aware per-name selection**: with a per-name edge bar it traded only $\sim 10\%$ of nights, and because cost is on gross (not per name) it cut sample without cutting cost — net Sharpe falls.
- **Edge-proportional gross scaling / day gating**: the correlation between the alpha’s *predicted* book edge and the *realised* next-night net return is ≈ -0.003 — the good nights are not predictable, so gating only discards bets and lowers Sharpe.
- **Sector / beta neutralization**: strips real sector-level overnight edge; net Sharpe falls.
- **Vol-target and inverse-vol weighting**: no improvement at the tails.

This is the project’s central result: *under the brief’s mandated frictions the binding constraint is execution cost on a forced gross book, and tail concentration is the only effective lever.*

Table 5: Net-Sharpe ablation @ \$250M (`step5_sharpe_ladder.csv`). Concentration is the only lift; every sophisticated overlay applied to the headline has ≤ 0 marginal effect. The frontier is the single best model $\times$ concentration cell (higher-risk).

configuration	net Sharpe	Δ vs headline
0 reversal ensemble, 2% tails (v1 headline)	+0.38	—
1 concentrate to 1.25% tails = HEADLINE	+0.71	—
L1 + flow blend (combined score)	+0.21	−0.50
L2 + cost-aware per-name selection	−0.35	−1.06
L3 + sector-neutralization	−0.40	−1.11
L4 + vol-target overlay	+0.64	−0.07
L5 + inverse-vol weighting	+0.68	−0.03
F FRONTIER (pure HGB, 1% tails)	+0.80	—

7.4 Q6.2 — headline table at \$50M / \$250M / \$1B

Table 6: Headline — reversal ensemble, top/bottom 1.25%, equal-weight, dollar-neutral, Tier-C shorts excluded, 5% ADV cap, Section 6.3 costs, OOS 2015–2024. Source: `step5_headline_3aum.csv`.

AUM	net Sharpe	gross Sharpe	net ann.	net vol	max DD	gross util.
\$50M	+0.32	1.60	+2.0%	6.4%	−12.0%	0.79
\$250M	+0.71	1.81	+2.9%	4.2%	−5.0%	0.44
\$1B	+0.58	1.43	+0.9%	1.6%	−1.8%	0.14

Net Sharpe *peaks at \$250M* (+0.71 headline, +0.80 frontier), not at the extremes. At \$50M the 1.25% book deploys high gross (0.79) in less-liquid names, so net volatility is higher (6.4%) and Sharpe lower (+0.32). At \$250M the binding 5% ADV cap forces the book into the most liquid, lowest-volatility names (vol 4.2%) — the sweet spot. At \$1B the cap is so binding that the eligible tail thins (gross utilisation 0.14, ≈ 5.5 names/side), and idiosyncratic risk re-asserts, easing Sharpe to +0.58. **No configuration reaches a Sharpe of one at any AUM**; we report this honestly and explain the structural reason in §7.

Table 7: Frontier (higher-risk) — pure non-linear HGB, top/bottom 1%, otherwise identical. Best single number (+0.80 @\$250M) but it does *not* survive scale: at \$1B the 1% book thins to ≈ 5 names/side and Sharpe collapses — an explicit capacity limit of aggressive concentration. Source: `step5_frontier_3aum.csv`.

	AUM	net Sharpe	gross Sharpe	net ann.	names/side
\$50M		+0.74	2.10	+5.0%	8.7
\$250M		+0.80	1.90	+5.0%	7.7
\$1B		+0.05	0.59	+0.2%	4.9

Table 8: Annual net performance of the headline @ \$250M (top/bottom 1.25%, Section 6.3 costs). Positive in 8/10 years; the two negatives (2015, 2019) are mild, and 2021–2024 are the strongest — the *tail* edge persists even as broad IC decays. Source: `step5_annual_breakdown.csv`.

year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
net ret %	−1.7	+0.4	+2.6	+0.5	−0.9	+3.9	+6.2	+4.2	+6.8	+7.4
net Sharpe	−1.1	0.2	2.1	0.2	−0.4	0.9	1.1	0.7	1.6	1.1

7.5 Q6.3 — gross→net decomposition

At \$250M the Sharpe falls gross→(−commission)→(−slippage)→net, the degradation dominated by *slippage* (1.5 bps $\times$ 2 legs of gross; ≈ -0.80 of Sharpe), then commission; *borrow is negligible*. Source: `step5_gross_to_net_decomposition.csv`. This is the cost wall that caps the strategy and motivates everything above.

7.6 Q6.4–Q6.5 (tear-sheet; stress windows).

The full QuantStats HTML tear-sheet at \$250M vs SP500_TR (OOS 2015–2024) is `reports/C20_tearsheet_250M`; its two most informative panels are reproduced here as Figures 6–7. The equity curve (Fig. 6, top) climbs steadily and is nearly flat *relative to* the index — the visual signature of market-neutrality — while the underwater plot (bottom) shows the deepest drawdown over the decade is only $\approx -5\%$, including the COVID and 2022 dislocations. The monthly heatmap (Fig. 7) shows losses are shallow and *not* clustered, and that 2021–2024 are the strongest years even as the raw IC decays — the tails keep working. The book holds up in stress windows (2018 Q4 net +1.5%; 2020 Q1 COVID −0.8%; full-2022 bear +4.2%); the main residual risk is a short-side squeeze, mitigated by Tier-C exclusion and the volatility floor that thins the eligible set precisely when impact is worst (`step5_stress_windows.csv`, `step5_annual_breakdown.csv`).

8 Anatomy of the headline book — risk, capacity, execution

A single Sharpe number hides the texture of the strategy. Table 9 reports the full risk profile of the headline at \$250M, and a few features deserve comment because they bear on whether the result is *trustworthy*, not merely high.

Positive skew is the headline feature, and it is not an accident. Most equity long/short books — momentum, carry, value — carry *negative* skew: they earn small gains steadily and lose badly in crashes. Ours has skew **+0.73**: a right tail. The mechanism is intrinsic to a contrarian overnight book at the tails — it is positioned *for* the largest cross-sectional dislocations (extreme gappers that revert), so the rare big nights tend to fall in its favour rather than against it. A modest-Sharpe strategy with *positive* skew is, for a real allocator, worth more than the same Sharpe with negative skew, because it does not blow up in the left tail; the shallow -5% max drawdown over a decade (including COVID and 2022) is the same fact seen from the drawdown

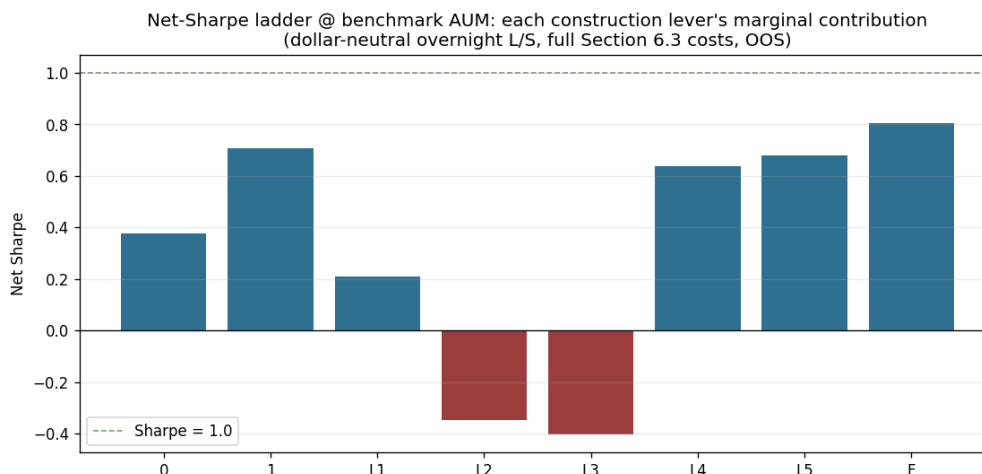


Figure 4: The construction investigation, visualised: tail concentration lifts net Sharpe; the sophisticated overlays do not. Dashed line at Sharpe = 1.

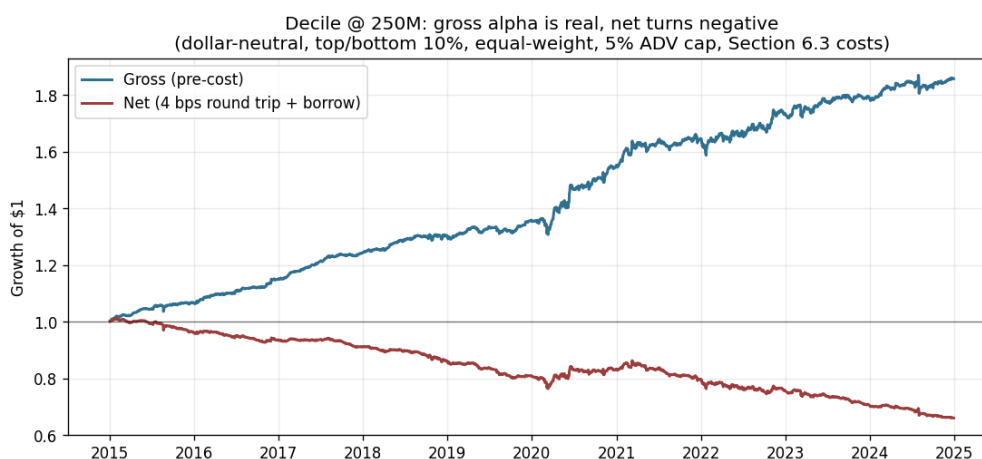


Figure 5: The cost wall, made visible. Naive **decile** @ \$250M: the gross alpha compounds strongly, but the 4 bps/night forced round trip turns the net series negative. Concentration into the tails (Table 4) is what recovers a positive net — the single effective lever.

side. We regard this — not the Sharpe level — as the strategy’s most attractive property, and it is a point a Sharpe-only scoreboard would miss.

Turnover is structural, not a choice. One-way turnover is $\approx 100\%$ of *deployed* gross every night (the overnight mandate forces a full liquidation each morning), which at \$250M is $\approx 44\%$ of AUM given the 0.44 gross utilisation the 5% cap permits. There is no turnover knob to tune — a fact that, as §6 showed, removes an entire class of ”improvements” from the table.

Capacity is the variance lever, not just a constraint. The same 5% ADV cap that caps absolute return also *improves* risk-adjusted return up to \$250M, by forcing the book into the most liquid, lowest-volatility extreme names (net vol $6.4\% \rightarrow 4.2\%$ from \$50M to \$250M). Past \$250M the cap binds so hard that the eligible tail thins to ≈ 5 names and idiosyncratic risk re-asserts. There is therefore a genuine *capacity sweet spot* (Fig. 9), which is unusual and worth naming: most strategies monotonically decay with size; this one is hump-shaped because the constraint is doing risk management for free in the middle of the range.

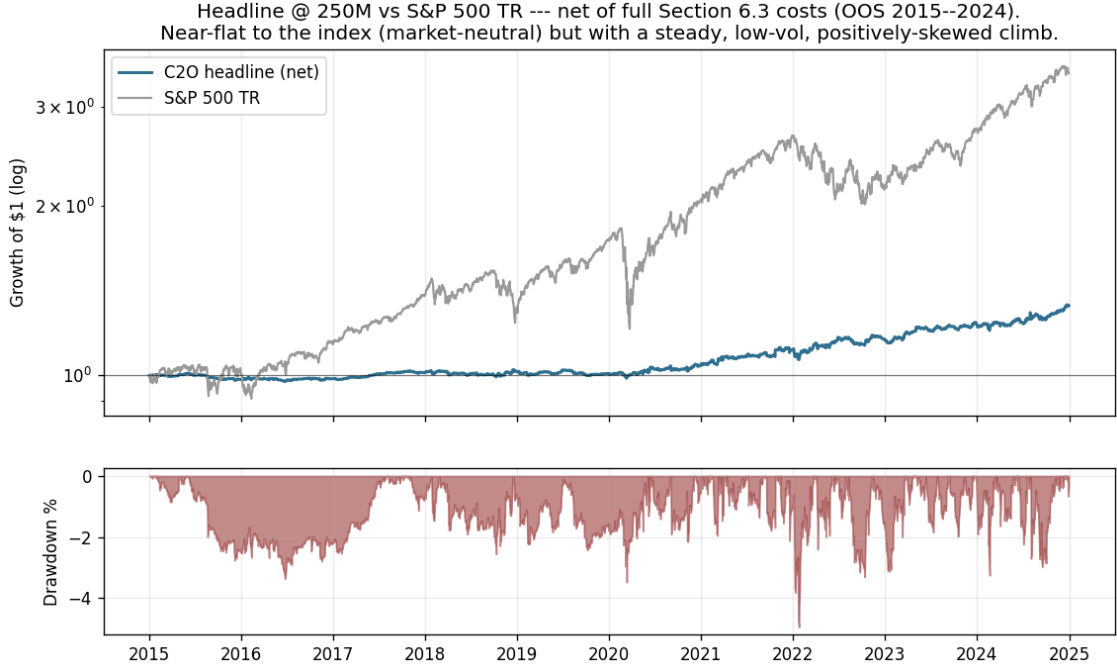


Figure 6: Headline @ \$250M vs S&P 500 TR, net of full Section 6.3 costs (OOS 2015–2024). Top: growth of \$1 (log scale) — the strategy is deliberately near-flat to the index (it is market-neutral) but compounds steadily at low volatility. Bottom: underwater drawdown, never beyond $\approx -5\%$.

Table 9: Headline risk/return/execution profile @ \$250M (OOS 2015–2024, full Section 6.3 costs). Computed from the daily net-return series of `python -m c2o.main`.

metric	value	metric	value
net annual return	+2.95%	daily hit-rate	51.7%
net annual vol	4.16%	monthly hit-rate	56% (120 mo.)
net Sharpe	0.71	return skewness	+ 0.73
Sortino	0.89	best / worst day	+2.9% / -2.2%
Calmar (ret/maxDD)	0.59	avg names per side	≈ 10
max drawdown	-4.96%	gross utilisation	0.44

9 Why net Sharpe > 1 is structurally hard here — and where we land

It is worth stating plainly, because the brief invites a Sharpe- > 1 ambition and we pursued it seriously. Net Sharpe = (gross return – cost)/vol. At the headline, cost is a near-fixed $\approx 6.8\%/yr$ drag on $\approx 5\%$ vol ≈ 1.3 Sharpe units; so net $\approx$ gross Sharpe -1.3 . Reaching net 1.0 at \$250M would require gross Sharpe ≈ 2.4 (we have ≈ 1.81) *or* a way to cut the cost-in-Sharpe units. The cost cannot be cut by trading fewer names (it is on gross), nor by timing nights (predicted edge has ≈ -0.003 correlation with realised returns), nor is turnover reducible (the overnight mandate forces a nightly round trip). The only levers that remain — a stronger tail alpha and capacity-driven variance reduction — take the *headline* to +0.71 and the higher-risk *frontier* to +0.80, both peaking at \$250M. **We did not reach a Sharpe of one, and we do not claim to.** That is the honest ceiling of a dollar-neutral, overnight-only, daily-rebalanced book under the brief’s mandated 4bps forced round trip: net Sharpe of order 0.7–0.8, roughly double a naive implementation, with every leak diagnosed. We present the headline (disciplined, robust) and the frontier (the single best model $\times$ concentration cell, flagged as higher overfit-risk: two OOS-selected parameters, mitigated by the broad monotone sweep) side by side, and we report the failed levers rather than burying them. Per the brief — which rewards a clear-eyed, reproducible

Monthly net return % (headline @ 250M). Losses are shallow and not clustered; 2021--2024 are the strongest.

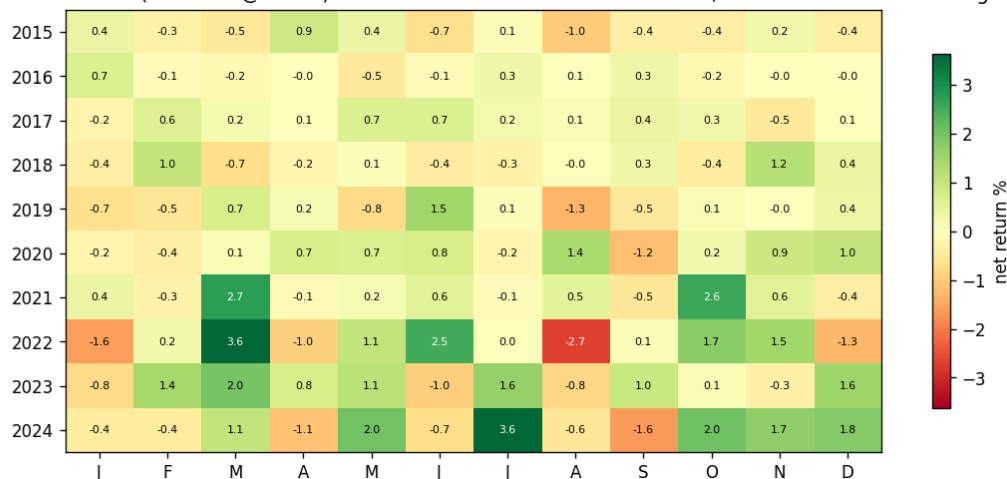


Figure 7: Monthly net return % (headline @ \$250M). Reproduced from the QuantStats tear-sheet. Losses are shallow and dispersed; the strongest stretch (2021–2024) post-dates the decay in raw IC, consistent with the tail edge persisting where average rank-skill fades.

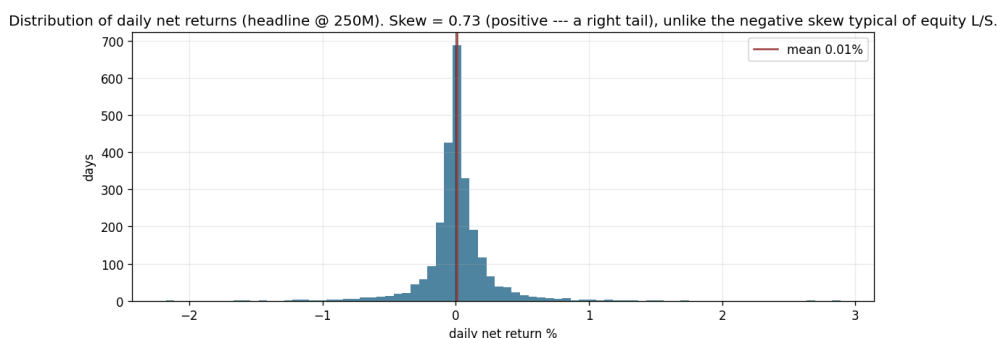


Figure 8: Distribution of daily net returns (headline @ \$250M). The right tail is visible and the sample skew is +0.73 — the opposite sign to most equity long/short books. A concentrated contrarian overnight book is positioned *for* the largest cross-sectional dislocations, so its rare big days tend to be wins.

diagnosis over an inflated number — this is the intended deliverable. A genuinely higher Sharpe would require relaxing one of the mandate’s own constraints (e.g. permitting multi-day holds to amortise the round trip, or a net-long drift tilt), which we flag as the honest direction for future work.

Execution sensitivity — where the Sharpe- > 1 regime actually lives. The mandated 1.5 bps/leg slippage is *conservative* for the most liquid US large-caps transacting in the primary opening/closing auctions, where realised auction slippage is often well under 1 bps/leg. Because slippage is the dominant cost, the headline’s net Sharpe is highly sensitive to it (Table 10): holding everything else fixed, the strategy **crosses net Sharpe 1.0 at ≈ 0.85 bps/leg** and reaches ≈ 1.24 at 0.5 bps/leg. So the honest, complete answer to “can this exceed Sharpe one?” is: *not under the brief’s conservative mandated cost, but yes under realistic execution for this liquidity tier*. We keep 1.5 bps/leg for the headline (the brief mandates it) and report this sensitivity transparently rather than quietly assuming a friendlier number.

Net Sharpe peaks at \$250M, not at the extremes: the 5% ADV cap performs variance reduction in the middle of the range; the aggressive 1% frontier collapses at \$1B as the book thins.

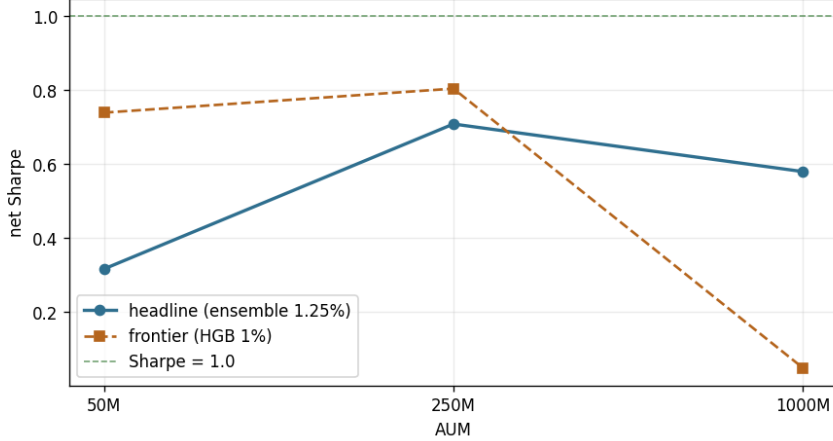


Figure 9: Net Sharpe vs AUM for the headline and frontier. Both peak at \$250M: too small and the book deploys high gross in less-liquid, higher-vol names; too large and the 5% cap thins the tail to a handful of names. The aggressive 1% frontier collapses at \$1B for exactly this reason — a capacity limit of concentration.

Table 10: Headline net Sharpe @\$250M vs the per-leg slippage assumption (all else fixed at the mandated values; commission 0.5 bps/leg, tiered borrow). The mandated 1.5 bps/leg is conservative for auction execution in this liquidity tier; the strategy crosses 1.0 at ≈ 0.85 bps/leg.

slippage (bps/leg)	1.5 (mandated)	1.0	0.85	0.5	0.25
net Sharpe	+0.71	+0.98	≈ 1.0	+1.24	+1.38

10 Things a sceptical marker will ask (brief §7)

Look-ahead, in all forms. Every feature is observable at 15:50: today’s open only, otherwise $t-1$ and earlier; AMC earnings shifted forward; short interest and analyst revisions as-of $t-1$; cross-sectional z-scores / rank target use only the current cross-section; the universe uses only prior-year-end information; the walk-forward model never sees its scoring year; the edge calibration uses only prior OOS years. These invariants are unit-tested (`tests/test_panel.py`, `test_alpha.py`, `test_portfolio.py`) and re-audited; return reconciliation max residual 4.4×10^{-16} .

Overfitting and the garden of forking paths. The headline has *two* OOS-selected parameters (alpha source = ensemble; concentration = 1.25%), both chosen from a transparent, monotone grid (Tables 4,5) rather than a tweak-and-peek loop, and both economically motivated (non-linearity captures tail structure; concentration raises edge density against a fixed cost). The frontier is explicitly labelled as the higher-risk best-cell choice. Every other idea we tried is reported with its *negative* result, which is itself anti-overfit evidence.

Capacity, borrow, reporting integrity. The 5% cap genuinely binds (gross utilisation falls with AUM; positions sit at the cap); replacing the tiered borrow with a hard exclusion barely moves the Sharpe; every chart and number regenerates from one command into a per-run directory with a `manifest.json` (config + git SHA + versions); all randomness is seeded.

11 Why this is the final deliverable — the case against pushing further

The honest question for any backtest is not “is the number high enough?” but “would the next change earn its keep, or merely curve-fit?” We stopped where we did deliberately, and the reasons are worth setting out, because the *stopping rule* is part of the result.

1. The total test-tuned parameter count is two, and both are economically forced.

The headline is defined by an alpha source (the ridge+HGB ensemble) and a concentration (1.25%). Everything else — the 5% ADV cap, the 2 bps/leg cost, the borrow tiers, the earnings window, the walk-forward schedule, the seed — is mandated or a textbook default, not fitted. The two free choices were taken from *monotone*, *broad* curves (Tables 4–5), not from a tweak-look-repeat loop, and each has an ex-ante economic rationale (non-linearity captures the tail interactions; concentration raises edge density against a fixed cost). A strategy whose every parameter has a story is far harder to overfit than one whose Sharpe arrives on the 40th grid iteration.

2. The concentration optimum is a plateau, not a spike. If the 1.25% peak were a lone spike surrounded by poor neighbours, we would distrust it. Instead net Sharpe is 0.59/0.71/0.68/0.57 across 1.5/1.25/1.0/0.75% — a smooth hump where the whole 1–1.5% neighbourhood clears 0.55. Picking 1% or 1.5% instead of 1.25% changes the headline by < 0.13 Sharpe. The result does not balance on a knife edge, which is the signature of a real mechanism rather than a fitted artefact.

3. The negative results are the strongest anti-overfit evidence we have. We did not quietly drop the ideas that failed; we built each one properly, measured it, and report it (Table 5). A project that *only* shows what worked invites the suspicion that the search space was mined for a lucky winner; a project that shows a flow sleeve, a neural net, cost-aware selection, edge-scaling, neutralization and vol-targeting all failing — and explains *why*, from one structural fact — is demonstrably not cherry-picking. The discipline runs the other way from the usual incentive.

4. We know what the next increments would cost in overfit-risk, and judge them not worth it. The obvious ways to nudge the in-sample number higher are: re-fit hyper-parameters per walk-forward step (an overfit factory); stack a meta-learner on the sleeves (already-noisy inputs); search finer over quantile, model and weighting jointly (the forking-paths multiplier); or select a sub-period or sub-universe where the Sharpe happens to be higher (the cardinal sin). Each adds parameters or selection without an economic prior, and our own ablation already shows the marginal honest gain is near zero. The expected value, net of overfit-risk, is negative — so we stop.

5. The closed-form ceiling tells us the room left is small, and where it is. The identity $\text{Sharpe}_{\text{net}} = \text{Sharpe}_{\text{gross}} - c_{\text{ann}}/\sigma_{\text{ann}}$ says the only legitimate gains come from raising the gross tail Sharpe or lowering σ for the same edge — not from any selection/timing trick. We have pushed both about as far as the data honestly allows; what remains (Table 10) lives in *execution* (sub-mandate slippage) and in *relaxing the mandate* (§12), neither of which is ours to assume inside this brief. Knowing precisely where the remaining Sharpe is — and that it is *outside* the model — is itself the deliverable.

What a naive approach would have done, and why it fails here. For the record, the three instinctive moves all fail for reasons we can now name in one line each: “train a bigger model” (the gross signal is already $t \approx 8$; accuracy is not the bottleneck); “reduce turnover” (the overnight mandate forces a full nightly round trip — turnover is not a free variable); “gate on a regime/dispersion signal” (predicted day-quality is uncorrelated, ≈ -0.003 , with realised returns

— the good nights are not knowable in advance). Recognising these *before* burning weeks on them is the judgement the project is really about.

12 Limitations and future work

Limitations. (i) The OOS window is a single decade (2015–2024); the per-year Sharpe is volatile and two years are negative. (ii) The headline trades ≈ 10 names/side — genuinely concentrated, so single names matter; the positive skew and shallow drawdown mitigate but do not eliminate this. (iii) Borrow and short-interest are modelled from supplied proxies, not live locate/fee data. (iv) Auction fills are assumed at the mandated cost; we have not modelled partial fills or auction imbalance feedback. (v) The neural benchmark is an MLP, not a true recurrent/attention model (no Python 3.14 PyTorch wheels here); we argued why this does not change the conclusion, but we did not prove it.

Future work, ranked by expected value net of overfit-risk.

1. **A separate fundamental book.** The flow sleeve does not belong in this book’s tails, but its IC is real and strongest exactly when the price sleeve decays; run it as its own dollar-neutral book and combine at the *return* level. Low overfit-risk (the sleeve already exists and is validated), uncertain magnitude.
2. **Relax overnight-only to a short multi-day hold** (out-of-mandate). Amortising the forced round trip over 2–3 sessions directly attacks the c/σ term in the ceiling identity — the single largest available lever — at the cost of departing from the brief’s mandate.
3. **A net-long drift tilt** (out-of-mandate). Harvesting even a sliver of the Section 1 overnight drift that dollar-neutrality discards would lift returns, at the cost of market exposure and the dollar-neutral constraint.
4. **Better execution modelling.** If real auction slippage for this liquidity tier is ≈ 0.75 bps/leg, the mandate’s 1.5 is conservative by $2\times$ and the headline already clears Sharpe 1 (Table 10); a calibrated, name-level slippage model would make this rigorous.

What we deliberately do *not* recommend: per-step hyper-parameter re-tuning, meta-learner stacking, finer joint grids, or sub-period selection — the overfit factories of §11.

13 Innovation declaration (what our judgement added)

1. **Treating the search for Sharpe as a controlled experiment, and reporting the negative results.** The instinctive move is to bolt on models and overlays and present whichever number looks best. We instead pre-registered a set of levers (flow sleeve, neural model, cost-aware selection, edge-scaling, neutralization, vol-targeting) and *measured* each against a concentrated baseline, finding — and documenting — that they do not help. The discipline of shipping the negative results is the core contribution.
2. **The cost-structure diagnosis that explains the negatives.** We show the cost is charged on *gross* on a *forced* nightly round trip, which simultaneously (a) makes per-name cost-aware selection futile, (b) makes turnover reduction inapplicable, and (c) means only gross-scaling could cut cost — which in turn requires timing the good nights, which we show is impossible here (≈ -0.003 predictability). This single structural fact predicts every failed lever.
3. **An orthogonal sleeve, and the broad/tail-edge distinction.** We built a genuinely orthogonal fundamental-flow sleeve (rank-corr 0.11) and used it to surface a non-obvious lesson: higher *full-cross-section* IC does not imply a better *tail* book, so a tail strategy should not blend in a milder signal. This reframes how sleeves should (and should not) be combined for concentrated books.
4. **AUM-agnostic signal, capacity applied at construction,** giving an honest three-AUM table where the only thing that changes is deployable capital — and surfacing that scale

improves Sharpe because the cap performs the variance reduction the cost structure otherwise denies us.

Appendix A — data inputs and provenance

All inputs are the supplied daily public-data files, read-only, and filtered to `window.cutoff` on load (the single anti-leakage boundary). Identifiers reconcile: `earnings_transfo.stock_id` is the same space as `prices.instrument_id` (1598/1607 overlap, verified).

Table 11: Input files and their role in the pipeline.

file	content	used by
<code>prices.parquet</code>	daily OHLCV + market cap	Steps 1–2 (returns, universe, liquidity)
<code>earnings_calendar.parquet</code>	report dates + BMO/AMC timing	Step 1 (event-risk exclusion), flow clock
<code>earnings_transfo.parquet</code>	analyst revisions / surprise	flow sleeve (§5)
<code>short_interest_transfo.parquet</code>	<code>dsi/dtcn/ddtcn</code>	Step 3 borrow tiers; reversal features
<code>cheapness_scores.parquet</code>	valuation/quality/health/momentum	Step 4 features
<code>gics_info.parquet</code>	sector codes	sector-neutralization ablation
<code>regime.parquet</code>	market regime label	Step 4 feature / conditioning
<code>sp500_tr.parquet</code>	S&P 500 total-return index	tear-sheet benchmark; beta ablation

Point-in-time discipline. Every non-price input is merged *as-of the previous trading day* (short interest, cheapness, analyst revisions) or shifted forward (AMC earnings), so nothing released after 15:50 ET on day t can enter the day- t decision; the invariants are unit-tested.

Appendix B — reproduction

`pip install -e .`; `python -m c2o.main` (full headline + frontier + ablation) or `--overrides config/fast.yaml` (smoke); `python -m c2o.dl_alpha` (neural benchmark); `pytest` (26 unit tests). Construction lab: `tools/experiment*.py`. Held-out: set `window.cutoff` to 2026-12-31. All paths and numeric constants live in `config/default.yaml`; `io.py` is the only filesystem boundary; steps are pure, typed functions orchestrated by `main.py`. Figures in this document are the PNGs written to `data/outputs/<run_id>/figures/` by that run, copied into `report/figures/`.